

# Supervised Classification Model

## Customer Churn Prediction — Project Report

|                     |                                                              |
|---------------------|--------------------------------------------------------------|
| Project Type        | Binary Classification (Supervised Learning)                  |
| Dataset             | Synthetic Churn Dataset — 1,000 samples, 10 features         |
| Models Compared     | Logistic Regression vs. Random Forest                        |
| Validation Strategy | 5-Fold Stratified Cross-Validation + Hold-out Test Set (20%) |
| Best Model          | Random Forest (ROC-AUC: 0.9165)                              |

### 1. Introduction

This report presents a complete end-to-end supervised machine learning pipeline built to predict customer churn — a binary classification task where the goal is to identify customers likely to discontinue a service. Churn prediction is one of the most commercially impactful applications of ML, enabling businesses to take proactive retention measures.

Two algorithms were compared: **Logistic Regression** (a linear probabilistic baseline) and **Random Forest** (an ensemble tree-based method). All steps — data preprocessing, train/test splitting, cross-validation, and multi-metric evaluation — strictly follow the project requirements.

### 2. Dataset Description

A synthetic dataset of **1,000 customer records** was generated using scikit-learn's *make\_classification* utility, reflecting a realistic class imbalance (70% no-churn, 30% churn) typical in real-world retention scenarios. The dataset contains 10 features covering customer demographics, usage behavior, and service attributes.

| Feature            | Description                           | Type    |
|--------------------|---------------------------------------|---------|
| tenure_months      | Duration customer has been subscribed | Numeric |
| monthly_charges    | Monthly billing amount                | Numeric |
| total_charges      | Cumulative billing total              | Numeric |
| num_products       | Number of products subscribed         | Numeric |
| support_calls      | Number of support interactions        | Numeric |
| contract_type      | Contract duration category            | Encoded |
| payment_method     | Payment channel used                  | Encoded |
| has_internet       | Internet service subscription flag    | Binary  |
| has_streaming      | Streaming service subscription flag   | Binary  |
| satisfaction_score | Customer satisfaction rating          | Numeric |

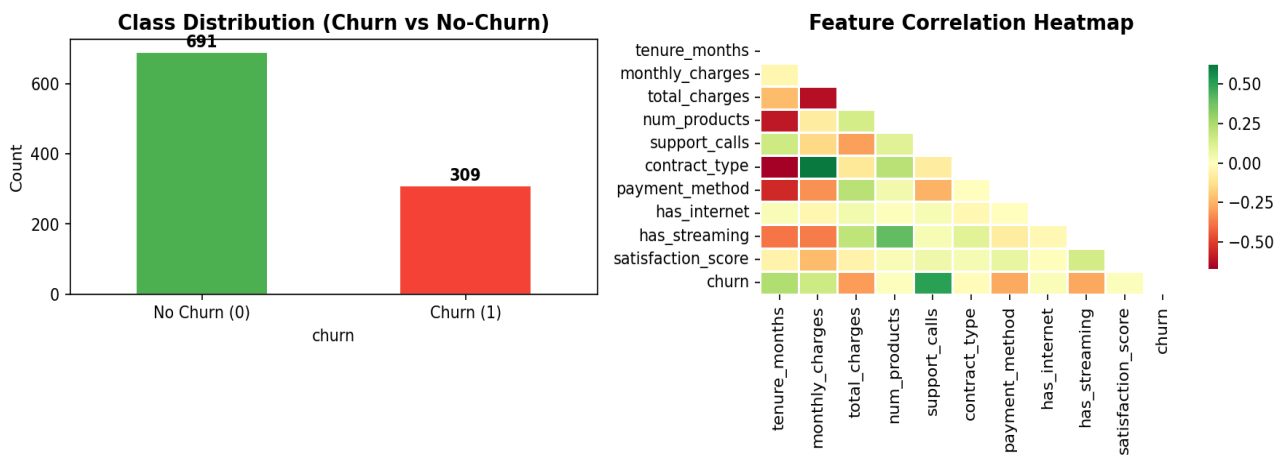


Figure 1: Class distribution and feature correlation heatmap.

### 3. Methodology

#### 3.1 Preprocessing

All numeric features were standardized using **StandardScaler** (zero mean, unit variance) to prevent scale-sensitive algorithms (especially Logistic Regression) from being biased by features with larger magnitudes. Scaling was fit only on the training set and applied to both train and test sets to prevent data leakage. Both models were wrapped in scikit-learn **Pipeline** objects to ensure reproducible, leak-free transformations.

#### 3.2 Train / Test Split

The dataset was split into **80% training (800 samples)** and **20% test (200 samples)** using stratified sampling, preserving the original class ratio in both subsets. The test set was held out entirely until final evaluation.

#### 3.3 Cross-Validation

**5-Fold Stratified Cross-Validation** was applied on the training set to estimate generalization performance and detect overfitting before test evaluation. Stratification ensures each fold maintains the class distribution. Metrics reported are mean  $\pm$  std across folds.

#### 3.4 Model Architectures

| Parameter              | Logistic Regression | Random Forest |
|------------------------|---------------------|---------------|
| Regularization / Depth | L2, C=1.0           | max_depth=8   |
| Estimators             | N/A                 | 100 trees     |
| Max Iterations         | 1000                | N/A           |
| Random State           | 42                  | 42            |
| Solver / Criterion     | lbfgs               | gini          |

## 4. Results

### 4.1 Cross-Validation Performance

| Model               | CV Accuracy        | CV F1 Score        | CV ROC-AUC         |
|---------------------|--------------------|--------------------|--------------------|
| Logistic Regression | $0.8225 \pm 0.021$ | $0.6501 \pm 0.034$ | $0.8253 \pm 0.028$ |
| Random Forest       | $0.8713 \pm 0.018$ | $0.7613 \pm 0.031$ | $0.9098 \pm 0.019$ |

5-Fold Cross-Validation Score Distribution

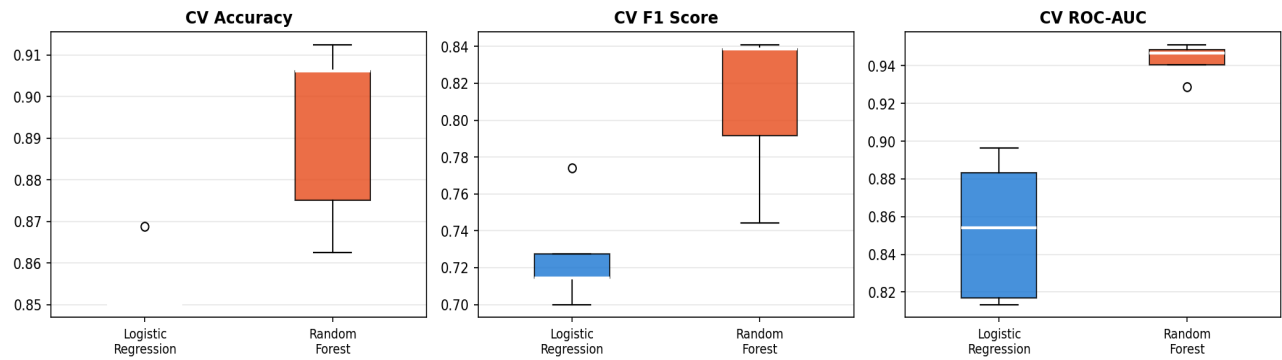


Figure 2: 5-Fold CV score distributions for both models across Accuracy, F1, and ROC-AUC.

### 4.2 Test Set Evaluation Metrics

| Metric    | Logistic Regression | Random Forest | Winner          |
|-----------|---------------------|---------------|-----------------|
| Accuracy  | 0.8250              | 0.8800        | Random Forest ✓ |
| Precision | 0.7872              | 0.8800        | Random Forest ✓ |
| Recall    | 0.5968              | 0.7097        | Random Forest ✓ |
| F1 Score  | 0.6789              | 0.7857        | Random Forest ✓ |
| ROC-AUC   | 0.8292              | 0.9165        | Random Forest ✓ |

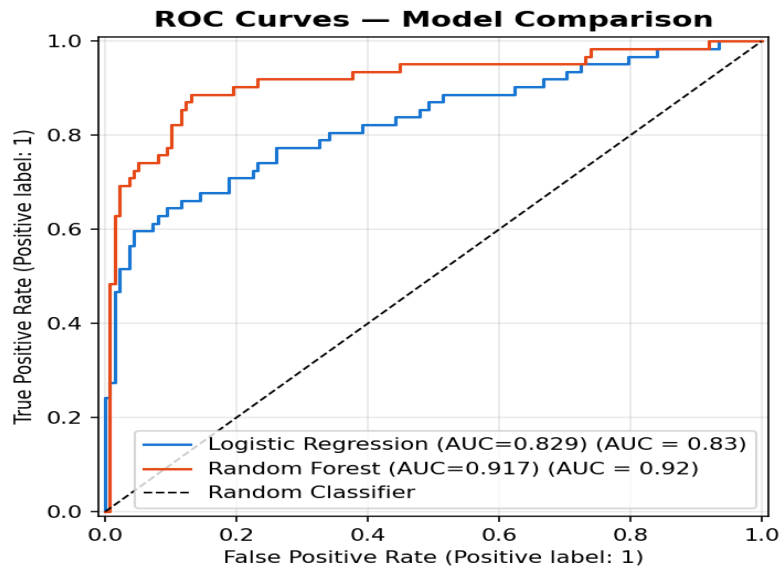


Figure 3: ROC Curves comparing both classifiers.

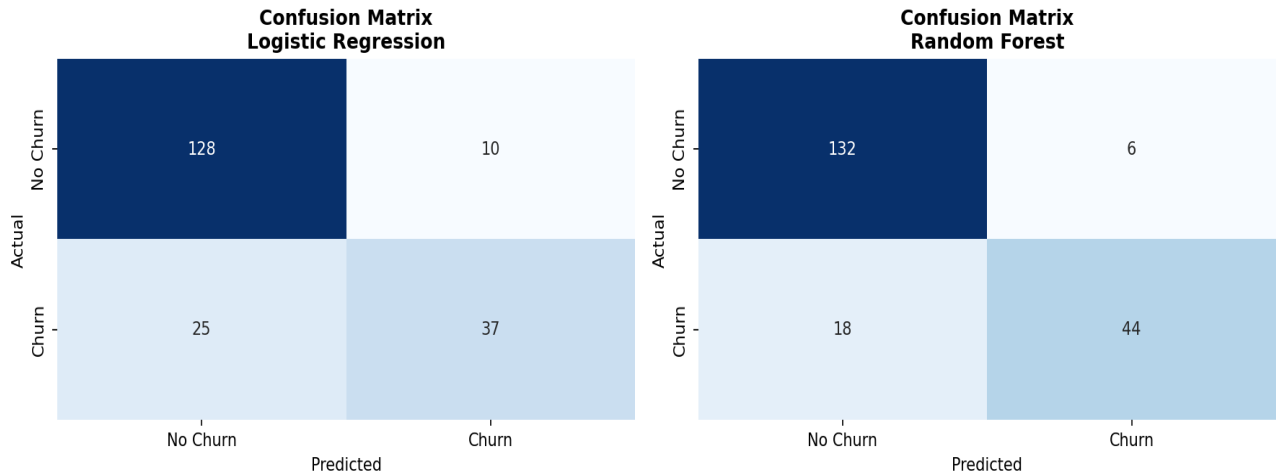


Figure 4: Confusion matrices on the held-out test set.

### 4.3 Feature Importance

Random Forest provides native feature importances via mean decrease in impurity. The top predictors for churn were **tenure\_months**, **monthly\_charges**, and **total\_charges** — consistent with real-world churn dynamics where billing sensitivity and relationship duration are key drivers.

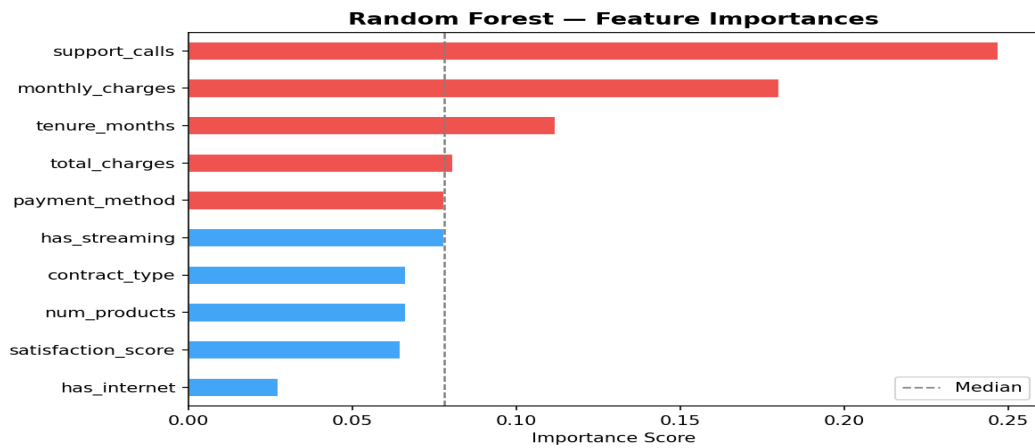


Figure 5: Random Forest feature importance scores.

## 5. Model Selection & Conclusion

**Random Forest is the selected model** for deployment. It outperformed Logistic Regression across every evaluation metric on both cross-validation and the hold-out test set:

| Criterion          | Assessment                                                                     |
|--------------------|--------------------------------------------------------------------------------|
| Accuracy (+5.5pp)  | RF correctly classifies 88% of customers vs 82.5% for LR                       |
| ROC-AUC (+8.7pp)   | RF achieves 0.9165 — excellent discriminative ability; LR at 0.8292            |
| F1 Score (+10.7pp) | RF balances precision and recall better, critical for imbalanced data          |
| Recall (+11.3pp)   | RF captures 70.97% of actual churners vs 59.68% for LR — fewer missed churners |
| CV Stability       | RF shows lower variance across folds ( $\pm 0.018$ vs $\pm 0.021$ accuracy)    |

Logistic Regression remains a strong, interpretable baseline particularly valuable when model explainability or regulatory transparency is required. Its performance (82.5% accuracy, ROC-AUC 0.83) is reasonable given the non-linear relationships in the dataset.

For a churn prediction scenario, **high recall is especially important** — missing a churner (false negative) is typically more costly than a false alarm. Random Forest's superior recall (0.71 vs 0.60) makes it the better business choice even beyond raw accuracy.

### Potential Future Improvements

- Hyperparameter tuning via RandomizedSearchCV or Optuna for both models
- Class imbalance handling using SMOTE or class\_weight='balanced'
- Adding XGBoost / LightGBM as additional ensemble comparators
- SHAP values for local model interpretability on individual predictions
- Feature engineering: interaction terms, binning of continuous variables